

🔗 AI WORKFLOW DESIGN

Trust Analytics *Workflow*

Making AI-generated analysis safe enough to act on.

Ask a business question → Validate the data path → Decide, review, or hand off

START WITH THE TRUST GAP →

AI can generate SQL and charts. People still need a reason to trust them.

CLIENT CONTEXT

A multi-asset trading platform is planning next month's promotion budget across crypto, gold, FX, GSS, and options.

"Which asset class should we prioritise?"

The AI can generate SQL and a chart showing crypto as the strongest completed-GTV campaign focus.

Which table?

Pending transactions?

Finance vs ops?

Without those checks, teams either over-trust the output or avoid using it.



WHAT AI CAN GENERATE

**SQL +
chart**

Fast, confident, and
useful as a first draft.

⚠️ **TRUST GAP**



NO PROOF PATH

WHAT THE BUSINESS NEEDS

**Ready for
decision**

Clear status: use it,
review it, or stop.

The gap is the missing proof between a generated output and a business decision.

GOAL Turn generated analysis into a clear status: use it, review it, or stop.

The ideal interface looks like one chatbox. The real workflow needs more structure.

The LLM is the engine. The workflow is the control system that makes its output usable.

IDEAL STATE

Ask once. The model infers everything.

Metric definition, source choice, SQL, data warnings, and business recommendation all arrive correctly in one response.

● AI ANALYST

Which asset class should we prioritise?

Prioritise crypto.

18.3B IDR completed GTV · 48% share

READY TO SEND

Leadership email draft

Recommend crypto as next month's primary growth lane.

PRODUCTION REALITY

Good answers still need control points.

The model may choose the wrong source, miss a warning, or sound certain when data definitions disagree.



Wrong table or grain



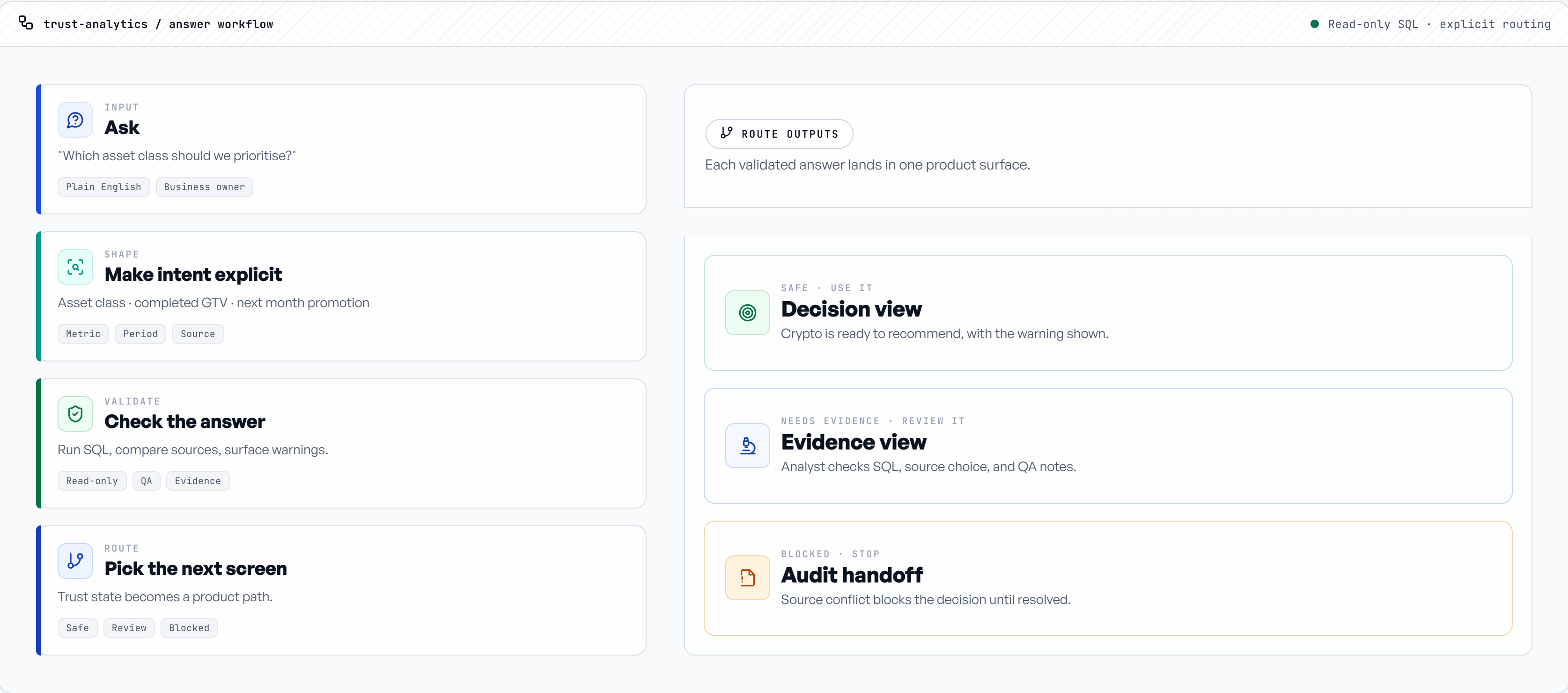
Source definition conflict



Hidden warning in a confident answer

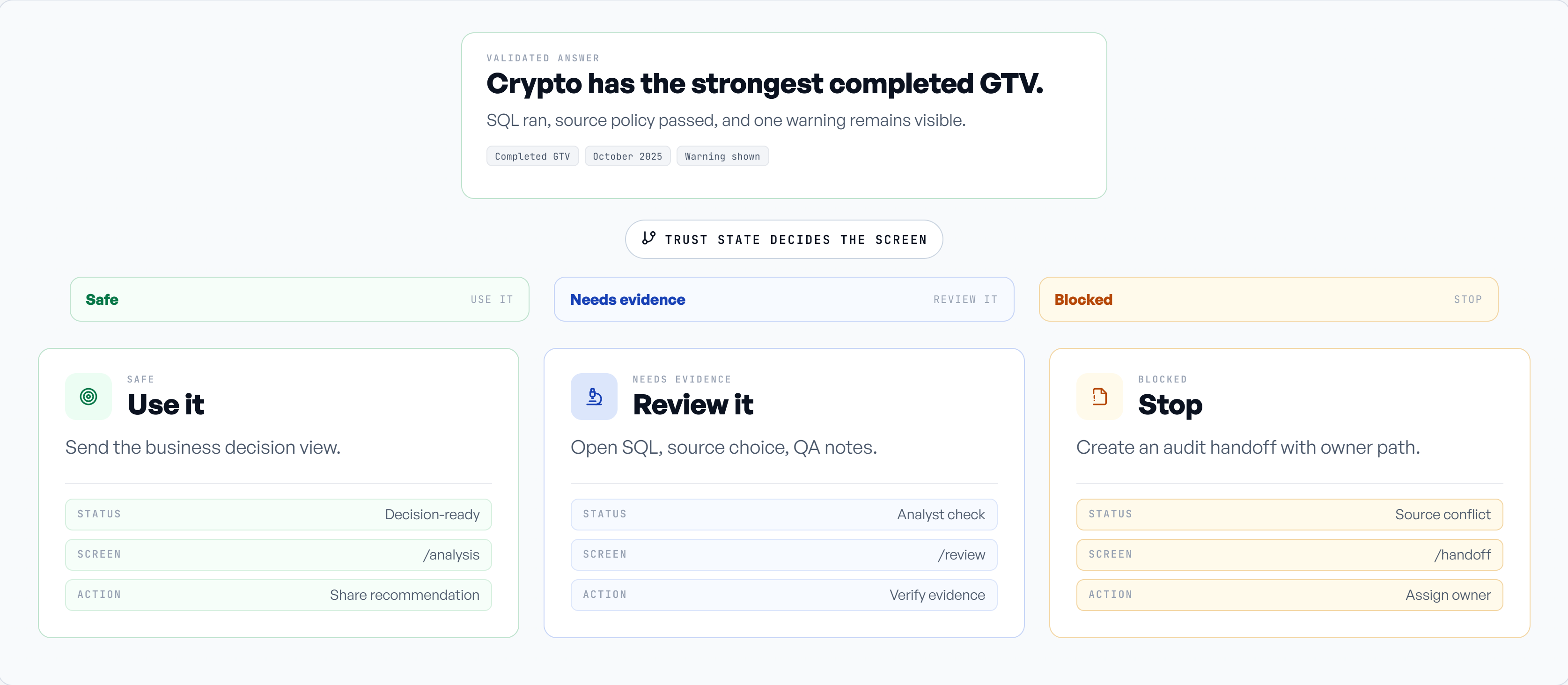
A workflow that turns analysis into action.

Every answer must carry a status, source evidence, and the next safe action.



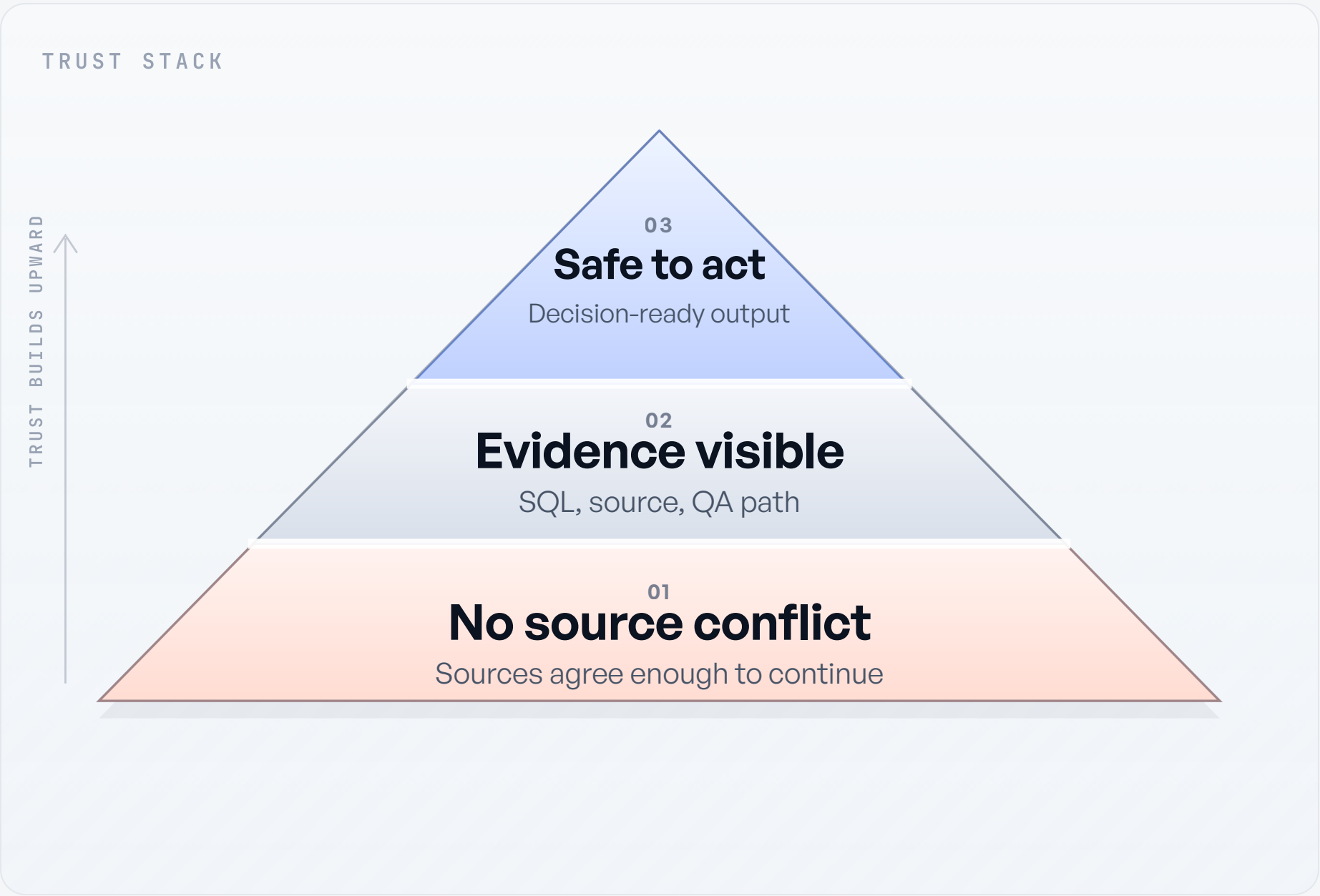
The workflow does not just answer. It decides where the answer belongs.

After validation, the product chooses a safe next screen instead of leaving trust as a feeling.



Trust state decides the screen.

Source conflict first. Evidence next. Action last. Each trust state routes the same answer to a different product surface.



ACTION

Business decision view

Plain-English conclusion, boundary, and source warning.

"Can I act on this?"



EVIDENCE

Analyst evidence view

Executed SQL, source rationale, QA, and challenge notes.

"Can I defend this?"



ESCALATION

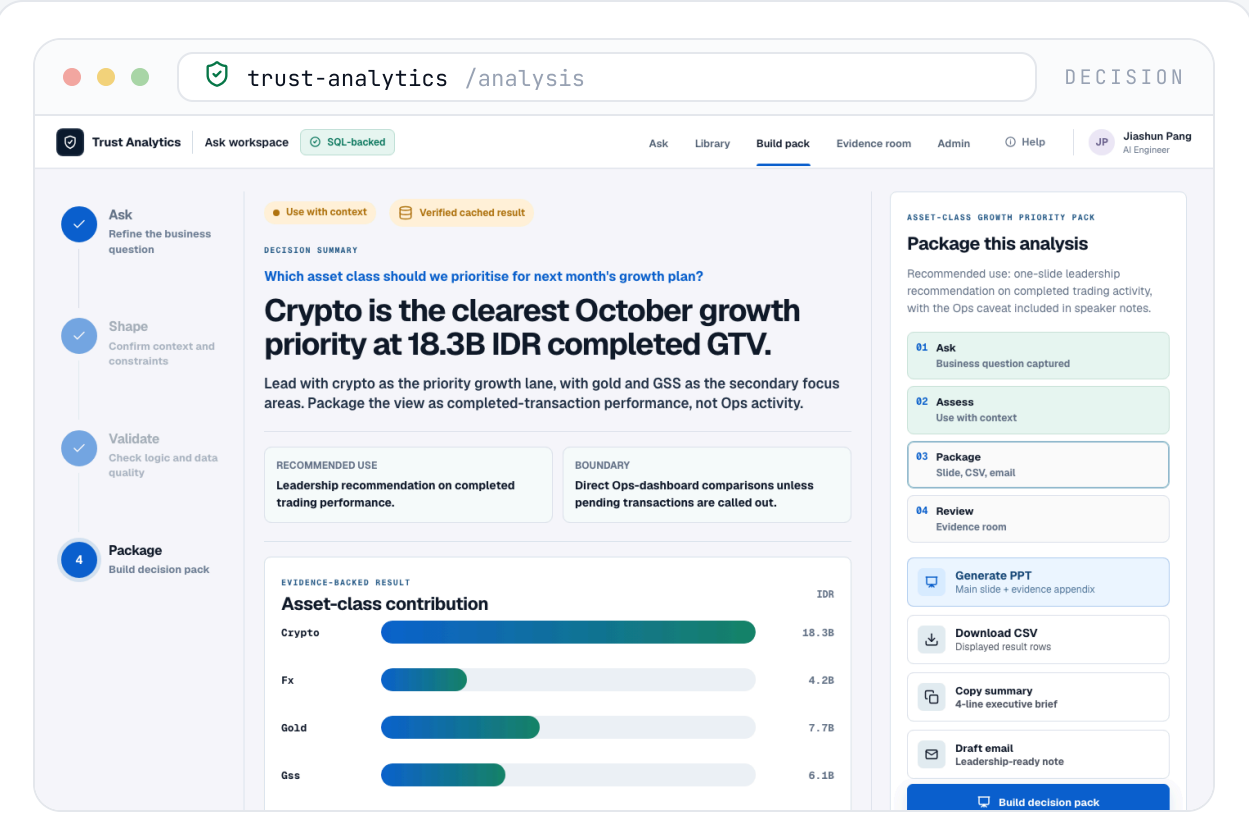
Audit handoff view

Named conflict, hold state, owner path, and resolution checklist.

"Should this be blocked?"

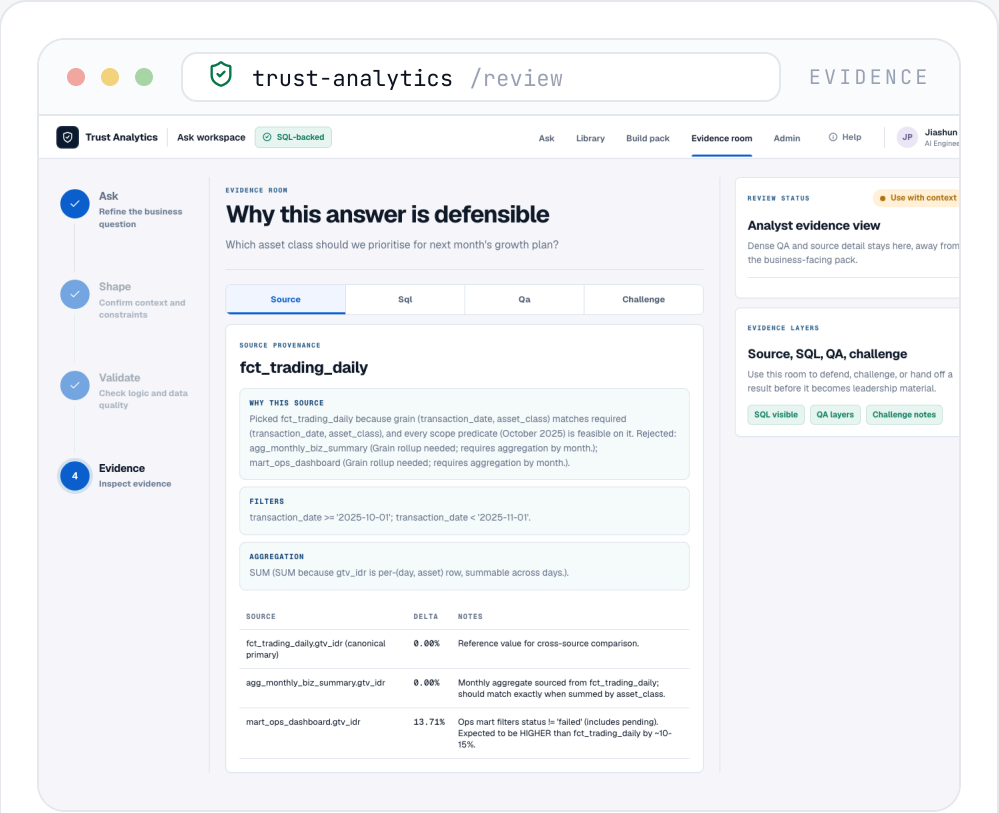
The workflow shows up as decisions, evidence, and escalation.

The user never receives a raw model answer. They receive a status and a next action.



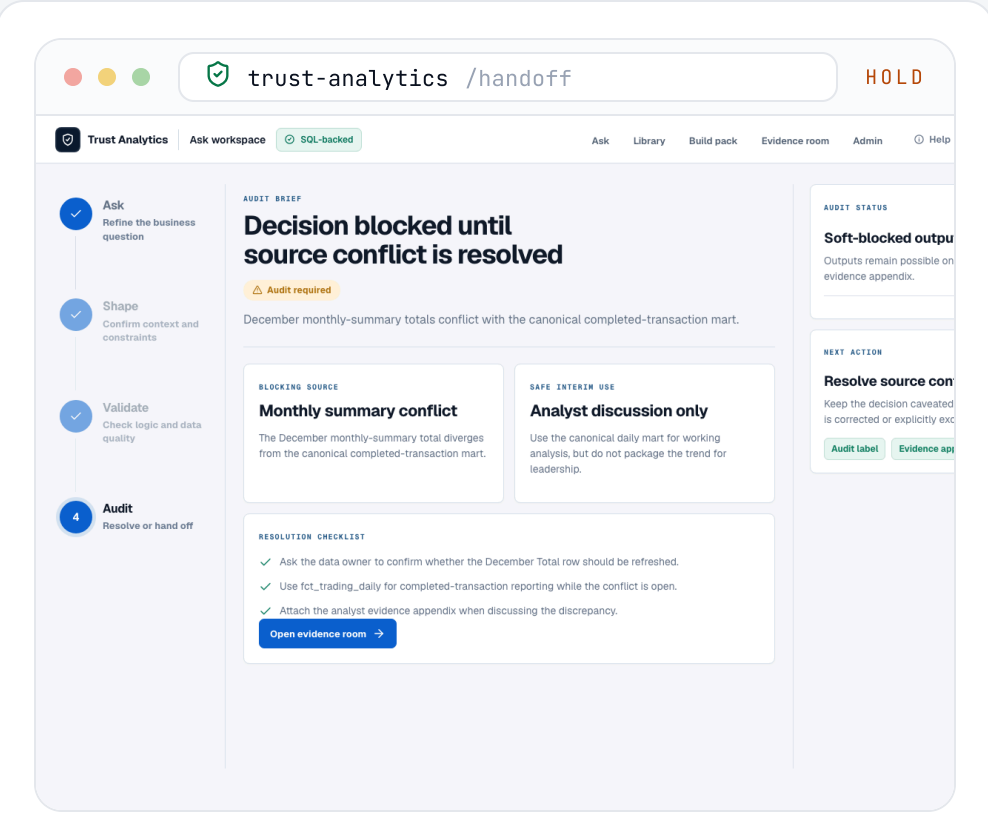
Decision-ready answer

Recommendation, boundary, warning, and pack actions sit on the same screen.



Defensible evidence

Source choice, SQL, reconciliation, and QA stay available for challenge.

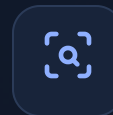


Escalation path

When sources disagree, the system blocks leadership packaging and names the next step.

A structured interface narrows the task and exposes uncertainty.

With the right workflow, even a smaller model can be useful because the product gives it clearer jobs and safer boundaries.



Question shaping

Turns vague intent into fields the model and SQL layer can use: objective, period, segment, dimension, output.

BETTER INPUT



SQL pre-flight

Checks whether the query matches the question before the result is trusted.

FEWER WRONG ANSWERS



Source reconciliation

Compares primary and alternative sources so definition conflicts are visible.

VISIBLE WARNINGS



Audit handoff

Stops the workflow when the data is not safe enough for a leadership decision.

SAFER ACTION

The product has to earn trust before the analysis can drive action.

1

Chat is powerful, but it is not the whole interface for trusted data work.

Free-form answers are fast. Decision workflows need explicit status, evidence, and warnings.

2

The workflow makes the model more useful.

Shaping, validation, and source checks reduce ambiguity before the user sees the answer.

3

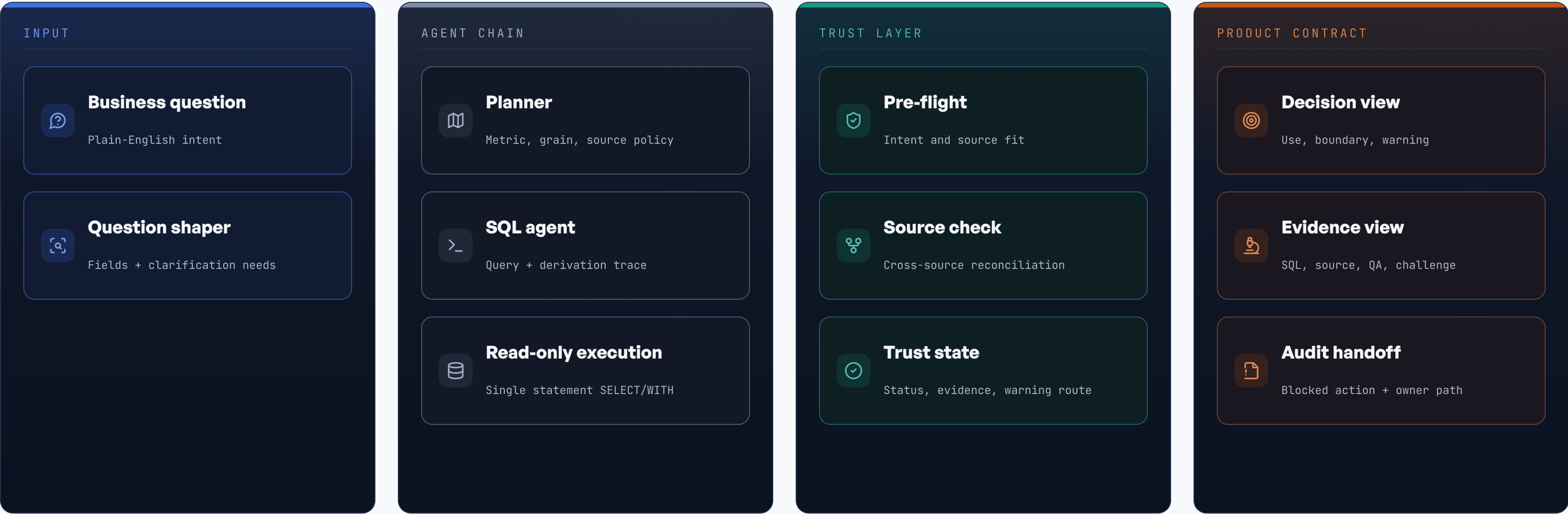
The workflow transfers beyond this fintech example.

The same pattern applies anywhere AI analysis needs to become a business decision: answer, evidence, action, escalation.

Appendix: implementation details and production guardrails.

Core architecture: model output is projected into a controlled product contract.

The UI does not read raw LLM text. It reads a structured analysis object with status, evidence, warnings, and exports.



Call logic: route every result to the safest next screen.

A live path is useful only when failure, cache, and audit states are explicit.

01

Shape request

POST /api/question/shape returns field quality, missing context, and recommended analysis path.

02

Run analysis

POST /api/runs starts planner, SQL agent, execution, pre-flight, and QA.

03

Route by status

Completed opens the decision pack. **Audit required** opens handoff. **Failed** never pretends to be verified.

04

Package output

CSV, email, and PPT outputs inherit the same warnings and evidence appendix.



No silent fallback.

If live validation is unavailable, the product can show a verified cached result only when it is labeled as cached and carries the original evidence.

Production guardrails that make the workflow trustworthy.

The implementation is intentionally narrow where trust matters and flexible where users need output.



Read-only SQL

Only SELECT/WITH queries pass. Multiple statements and mutating keywords are blocked.

```
sql_runner.validate_read_only_sql
```



Pre-flight validation

Checks source, filters, aggregation, and answer shape before QA sees the result.

```
pre_flight.py
```



Source reconciliation

Compares canonical and alternative sources, then names the disagreement instead of hiding it.

```
layer_b.py
```



Audit-first failure mode

When the answer is not safe enough, the user gets a handoff path rather than a confident guess.

```
workflow.py → audit_handoff
```



Decision pack export

PPT, CSV, and email outputs carry the same source warning and evidence trail.

```
portal.make_pptx
```



Operational telemetry

API calls and export events are tracked so the workflow can be improved with usage evidence.

```
telemetry.record_event
```